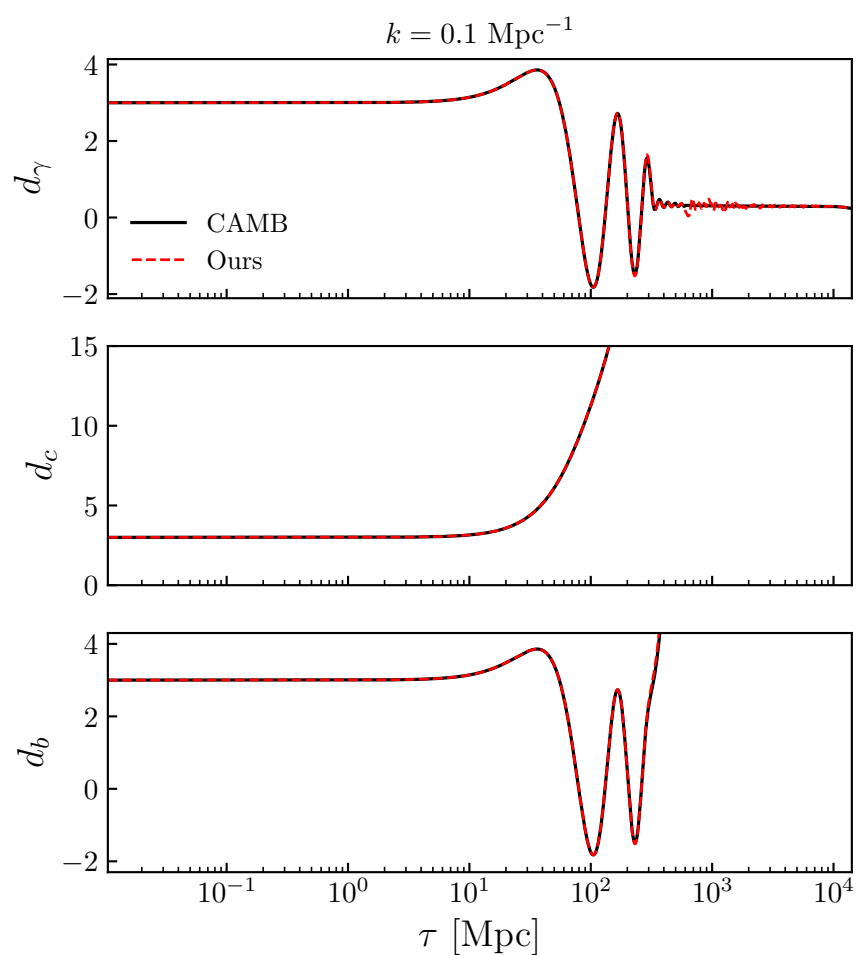


Boltzmann Code - Perturbations

Last week we evolved a single Fourier mode of a radiation-only universe. This week we will extend the same machinery to the full Λ CDM perturbation system. In particular, we'll add in CDM and baryons and couple the photons and baryons through Thomson scattering. By the end of this lab you will have a servicable Einstein-Boltzmann solver.



Setup: We continue to work in conformal Newtonian gauge with metric

$$ds^2 = a^2(\tau) [-(1 + 2\Phi) d\tau^2 + (1 - 2\Psi) d\vec{r}^2],$$

track the evolution of the perturbation $d_a \equiv \delta_a/(1 + w_a) - 3\Psi$ for each species, together with the velocity potential u_a , anisotropic-stress potential π_a , and the higher- l multipole tower $d_{l,a}$ for collisionless species. Remember that $d_{0,\nu} = d_\nu$, $d_{1,\nu} = u_\nu$, $d_{2,\nu} = (3/2)\pi_\nu$. The same multipole structure we had only for neutrinos last time will now also apply to photons after we leave the tight-coupling regime.

The new species are **CDM** and **baryons**. In the regime of small perturbations, we only need to track (d_c, u_c, d_b, u_b) . Instead of radiation domination like we had in **nu_phase_shift.pdf**, we will use our full Planck Λ CDM background cosmology which we implemented in **background.pdf**, with the ionization history $x_e(z)$ from **recombination.pdf** and **reionization.pdf**.

The metric perturbations are still determined at every time step directly from species perturbations (now with a ranging over all four species):

$$\Psi = -\frac{\gamma_{\text{tot}} (d_{\text{tot}} + 3\mathcal{H} u_{\text{tot}})}{k^2 + 3\gamma_{\text{tot}}}, \quad \Phi = \Psi - 3\gamma_{\text{tot}} \pi_{\text{tot}}, \quad (1)$$

where $\gamma_a \equiv 4\pi G a^2 (\bar{\rho}_a + \bar{P}_a)$, $\gamma_{\text{tot}} = \sum_a \gamma_a$, and the totals are weighted averages

$$d_{\text{tot}} = \sum_a x_a d_a, \quad u_{\text{tot}} = \sum_a x_a u_a, \quad \pi_{\text{tot}} = \sum_a x_a \pi_a, \quad x_a \equiv \frac{\gamma_a}{\gamma_{\text{tot}}}.$$

For CDM and baryons we have $\bar{P}_a = 0$ so $\gamma_a = 4\pi G a^2 \bar{\rho}_a$.

Part 1: Extend last week's **metric_perturbations** to take all four species as input and return (Φ, Ψ) using Eq. (1).

State vector: Following the same conventions as last week, the full state vector is

$$y = \left(\underbrace{d_c, u_c}_{\text{CDM}}, \underbrace{d_b, u_b}_{\text{baryon}}, \underbrace{d_\gamma, u_\gamma, \pi_\gamma, d_{3,\gamma}, \dots, d_{\text{LMAX}_\gamma, \gamma}}_{\text{photon}}, \underbrace{d_\nu, u_\nu, \pi_\nu, d_{3,\nu}, \dots, d_{\text{LMAX}_\nu, \nu}}_{\text{neutrino}} \right).$$

We take $\text{LMAX}_\gamma = \text{LMAX}_\nu = 15$ for this lab.

Initial conditions: We start integration deep in the radiation era, when the mode is still frozen on superhorizon scales. The initial condition for adiabatic growing modes are the same as last week:

$$d_a = -3\zeta, \quad u_a = \frac{5\tau_{\text{init}} \zeta}{15 + 4R_\nu}, \quad \pi_\nu = \frac{2\tau_{\text{init}}^2 \zeta}{3(15 + 4R_\nu)}, \quad (2)$$

with all higher- l neutrino multipoles vanishingly small. We set $\zeta = -1$ to match convention of the transfer functions that are easy to pull out of **CAMB**.

Part 2: Write `initial_state(tau_init, k, bg)` that returns the initial condition for the full state vector using Eq. (2). This is essentially last week's `initial_state` with two extra blocks for CDM and baryons.

Tight-coupling approximation: Before recombination, the rate at which CMB photons Thomson-scatter off free electrons is very fast. So, the Thomson scattering terms in the photon and baryon Euler equations,

$$u'_\gamma \supseteq \sigma_T x_e (u_b - u_\gamma), \quad u'_b \supseteq (1/R_b) \sigma_T x_e (u_\gamma - u_b), \quad R_b \equiv \frac{3\bar{\rho}_b}{4\bar{\rho}_\gamma},$$

are huge relative to other relevant time scales. This gives us a nasty stiff system. To get around this, we can use the tight-coupling approximation (TCA) where we evolve the combined photon-baryon fluid as a single entity while keeping a few low-order corrections in the photon mean-free path $\tau_c \equiv 1/(\sigma_T n_e)$.

The TCA equations for the photon-baryon fluid are

$$d'_\gamma = -k^2 u_\gamma, \quad u'_\gamma = -\mathcal{F} u_\gamma + \frac{d_\gamma}{3(1+R_b)} + \Phi + \frac{\Psi}{1+R_b}, \quad (3)$$

where the friction coefficient

$$\mathcal{F} \equiv \frac{\mathcal{H}R_b}{1+R_b} + 2k^2\tau_d, \quad \tau_d \equiv \frac{\tau_c}{6} \left[1 - \frac{14}{15(1+R_b)} + \frac{1}{(1+R_b)^2} \right], \quad (4)$$

contains two physically distinct pieces. The $\mathcal{H}R_b/(1+R_b)$ term is the friction from the baryon mass-loading on the photon fluid: in the limit $R_b \rightarrow 0$ (no baryons) it vanishes and the photons oscillate freely with sound speed $c_s^2 = 1/3$. For nonzero R_b the baryons drag, the sound speed slows to $c_s^2 = 1/(3(1+R_b))$, and the fluid loses kinetic energy to expansion. The $2k^2\tau_d$ piece is **Silk damping**, which quantifies the small slip between photons and baryons from photon diffusion. See [lecture_notes/section06.pdf](#) for more discussion.

The baryons are locked to the photons,

$$d'_b = -k^2 u_\gamma, \quad u'_b = u'_\gamma. \quad (5)$$

CDM is collisionless throughout, so its equations are the same in TCA and the full system:

$$d'_c = -k^2 u_c, \quad u'_c = -\mathcal{H} u_c + \Phi. \quad (6)$$

Note that CDM is non-relativistic, so only Φ drives the Euler equation. You can put this in contrast with the radiation Euler equation where both Φ and Ψ appear as sources.

The massless neutrinos are also collisionless and are evolved using exactly the same equations as last week. Reproducing them here for completeness: the monopole/dipole/quadrupole are

$$d'_\nu = -k^2 u_\nu, \quad u'_\nu = \frac{1}{3} d_\nu - k^2 \pi_\nu + \Phi + \Psi, \quad \pi'_\nu = \frac{4}{15} u_\nu - \frac{2}{5} k^2 d_{3,\nu}, \quad (7)$$

the higher multipoles ($l \geq 3$) obey the recursion

$$d'_{l,\nu} = \frac{l}{2l+1} d_{l-1,\nu} - \frac{l+1}{2l+1} k^2 d_{l+1,\nu}, \quad l \geq 3, \quad (8)$$

with $d_{2,\nu} = (3/2)\pi_\nu$ at $l = 3$, and the streaming closure at $l = \mathbf{LMAX}_\nu$ is

$$d'_{\mathbf{LMAX}_\nu,\nu} = d_{\mathbf{LMAX}_\nu-1,\nu} - \frac{\mathbf{LMAX}_\nu + 1}{\tau} d_{\mathbf{LMAX}_\nu,\nu}. \quad (9)$$

Part 3: Write `tca_rhs(tau, y, k, bg)` that assembles the time derivatives in TCA using Eqs. (3), (5), (6), and the neutrino block Eqs. (7), (8), (9). You will need helpers `scale_factor_from_tau(tau)` and a Thomson-rate σ_{Tan_e} as a function of τ . With this, you can now integrate the TCA system with `scipy.integrate.solve_ivp` starting from $\tau_{\text{init}} = 10^{-3}/k$.

Validation – TCA: Before moving on, let’s see how the TCA solution holds up against **CAMB**. We will compare against **CAMB**’s Newtonian-gauge perturbations using `camb.symbolic` to access δ_γ , q_γ , δ_c , δ_b , δ_ν , and Φ_N . Note that Φ_N from CAMB is our Ψ . Translating to our variables,

$$d_\gamma = \frac{3}{4}\delta_\gamma - 3\Psi, \quad u_\gamma = \frac{3}{4}q_\gamma/k, \quad d_c = \delta_c - 3\Psi, \quad d_b = \delta_b - 3\Psi, \quad d_\nu = \frac{3}{4}\delta_\nu - 3\Psi.$$

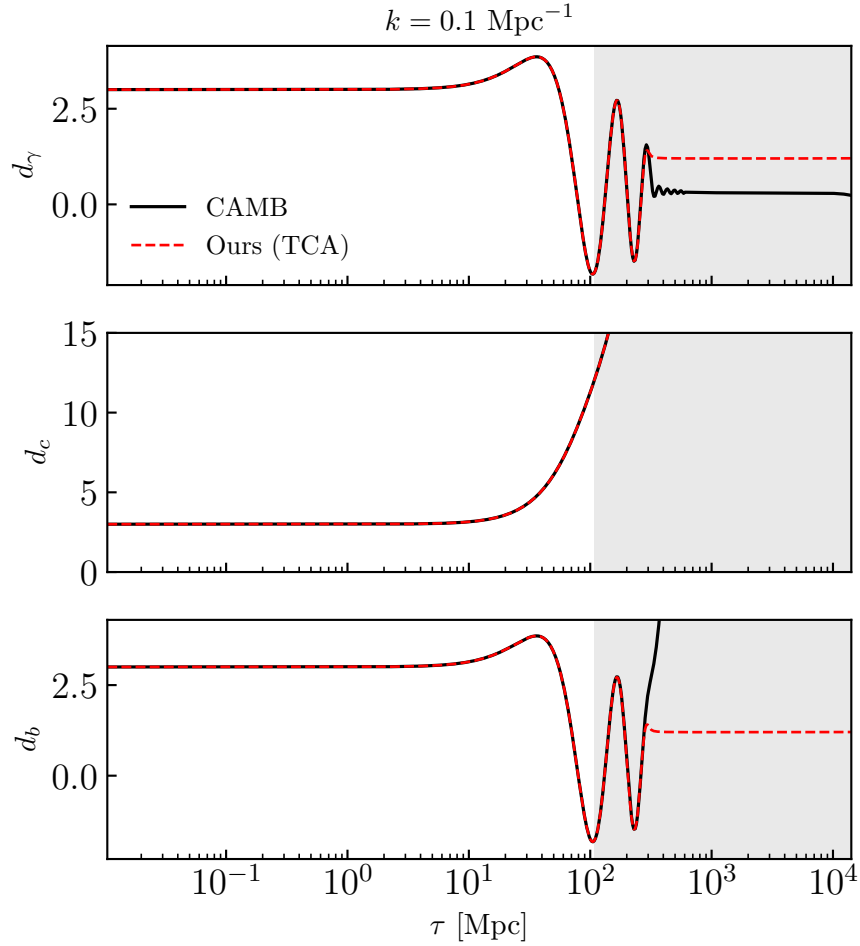
In code this look roughly like

```

1 import camb
2 from camb.symbolic import Delta_g, Delta_c, Delta_b, Delta_r, q_g, Phi_N
3
4 pars = camb.set_params(...)
5 results = camb.get_results(pars)
6
7
8 def camb_perts(k, tau):
9     out = results.get_time_evolution(
10         k, tau,
11         vars=[Delta_g, q_g, Delta_c, Delta_b, Delta_r, Phi_N],
12         frame='Newtonian',
13     )
14     delta_g, q_g_c, delta_c, delta_b, delta_r, phi_n = out.T
15     return {
16         'd_g': 0.75 * delta_g - 3.0 * phi_n,
17         'u_g': 0.75 * q_g_c / k,
18         'd_c': delta_c - 3.0 * phi_n,
19         'd_b': delta_b - 3.0 * phi_n,
20         'd_n': 0.75 * delta_r - 3.0 * phi_n,
21     }

```

For $k = 0.1 \text{ Mpc}^{-1}$ you should get something like



Our d_γ , d_c , and d_b agree with **CAMB** well at early times, but diverge once we pass into the regime where the tight-coupling approximation fails (the grey shaded region). This regime we define as $\tau > \tau_{\text{switch}}$ with $k\tau_c(\tau_{\text{switch}}) \simeq 0.02$. Basically, we say that once a photon's mean-free-path approaches a tenth of a wavelength, the photon-baryon plasma can no longer be treated as a single fluid.

Full system: Once $\tau > \tau_{\text{switch}}$, we keep the Thomson rate σ_{Tan_e} explicitly in the equations rather than expanding around $\sigma_{Tan_e} \rightarrow \infty$. We'll now track Photons and baryon perturbations separately.

Baryons. The baryon continuity equation is the same as before. The Euler equation gains a Thomson drag from the photons:

$$d'_b = -k^2 u_b, \quad u'_b = -\mathcal{H} u_b + \Phi + \frac{\sigma_{Tan_e}}{R_b} (u_\gamma - u_b). \quad (10)$$

The factor of $1/R_b$ comes from momentum conservation, the momentum transferred to baryons is the momentum lost by photons.

Photons. The photon continuity, Euler, and quadrupole equations are

$$\begin{aligned} d'_\gamma &= -k^2 u_\gamma, \\ u'_\gamma &= \frac{1}{3} d_\gamma - k^2 \pi_\gamma + \Phi + \Psi + \sigma_{Tan_e} (u_b - u_\gamma), \\ \pi'_\gamma &= \frac{4}{15} u_\gamma - \frac{2}{5} k^2 d_{3,\gamma} - \frac{9}{10} \sigma_{Tan_e} \pi_\gamma. \end{aligned} \quad (11)$$

Two changes from the neutrino case: (1) the Thomson term $\sigma_{Tan_e} (u_b - u_\gamma)$ in the Euler equation and (2) the Thomson damping $-\sigma_{Tan_e} \pi_\gamma$ in the quadrupole equation.

The higher- l photon multipoles obey the same recursion as neutrinos but with an additional Thomson damping term that kicks in for all $l \geq 3$:

$$d'_{l,\gamma} = \frac{l}{2l+1} d_{l-1,\gamma} - \frac{l+1}{2l+1} k^2 d_{l+1,\gamma} - \sigma_{Tan_e} d_{l,\gamma}, \quad l \geq 3, \quad (12)$$

with $d_{2,\gamma} = (3/2)\pi_\gamma$ used at $l = 3$. The streaming closure at $l = \mathbf{LMAX}_\gamma$ inherits

the same Thomson damping,

$$d'_{\text{LMAX}_\gamma, \gamma} = d_{\text{LMAX}_\gamma - 1, \gamma} - \frac{\text{LMAX}_\gamma + 1}{\tau} d_{\text{LMAX}_\gamma, \gamma} - \sigma_T n_e d_{\text{LMAX}_\gamma, \gamma}. \quad (13)$$

The CDM and neutrino equations are unchanged from TCA.

Part 4: Write `full_rhs(tau, y, k, bg)` that assembles the time derivatives for the full system using Eqs. (10), (11), (12), (13), and the unchanged CDM and neutrino blocks. Then write `solve_perturbations(k)` that (i) finds τ_{switch} from $k\tau_c \simeq 0.02$, (ii) integrates `tca_rhs` from τ_{init} to τ_{switch} , and (iii) hands the final TCA state to `full_rhs` and integrates it the rest of the way to today.

Validation – full system: Solve the full system for a bunch of different k . For $k = 0.1 \text{ Mpc}^{-1}$, you should reproduce the plot on the first page.

References:

- **Bashinsky & Seljak (2003)**, *Signatures of relativistic neutrinos in CMB anisotropy and matter clustering*, [astro-ph/0310198](#).
Like last week, this is the paper whose notation we're following.
- **Ma & Bertschinger (1995)**, *Cosmological Perturbation Theory in the Synchronous and Conformal Newtonian Gauges*, [astro-ph/9506072](#).
The standard reference for linear perturbations.